



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2022 Year 12 Course Assessment Task 2

Wednesday March 16, 2022

General instructions

- Working time – 90 minutes.
(plus 5 minutes reading time)
- Write using a black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- All necessary working should be shown in every question.
- Marks may be deducted for illegible or incomplete working.
- A NESA Reference Sheet is provided.

SECTION I

- Mark your answers on the answer grid provided (on page 3)

SECTION II

- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Ms Ham

☐ 12MXX.2 – Mr Sekaran

☐ 12MXX.3 – Mr Ho

Marker's use only.

QUESTION	1-5	6	7	8	9	10	Total	%
MARKS	5	15	11	7	7	9	54	

Section I

5 marks

Attempt Question 1 to 5

Allow approximately 6 minutes for this section

Mark your answers on the answer grid provided (labelled as page 3).

Glossary

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ – set of all natural numbers
- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – set of all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers
- $\mathbb{C}$ – set of all complex numbers

Questions

Marks

1. Which of the following are the principal arguments of the solutions to the equation 1

$$z^2 = 1 + i$$

(A) $\frac{\pi}{8}$ and $\frac{9\pi}{8}$

(C) $-\frac{7\pi}{8}$ and $\frac{\pi}{8}$

(B) $-\frac{\pi}{8}$ and $\frac{7\pi}{8}$

(D) $\frac{7\pi}{8}$ and $\frac{15\pi}{8}$

2. If $z = 3 - 4i$, which of the following is equal to $\frac{1}{1 - z}$? 1

(A) $\frac{-1 - 2i}{10}$

(B) $\frac{-1 + 2i}{10}$

(C) $\frac{-1 - i}{6}$

(D) $\frac{-1 + i}{6}$

3. Given $z_1 = \cos \alpha + i \sin \alpha$ and $z_2 = \cos \beta + i \sin \beta$, what is the value of 1

$$\arg \left(\frac{z_1 z_2}{z_1 + z_2} \right)$$

(A) $\frac{1}{2}\alpha\beta$

(B) $\frac{1}{2}(\alpha + \beta)$

(C) $\alpha\beta$

(D) $\alpha + \beta$

4. Given $x \in \mathbb{N}$ and $y \in \mathbb{N}$, which of the following statements is **false**? 1

(A) $\forall x \exists y (x - y = 1)$

(C) $\forall x \exists y (x - 2y = 0)$

(B) $\forall x \exists y (2x - y = 0)$

(D) $\exists x \exists y (x + y = 2)$

5. What is the negation of the statement:

1

$$n < -2 \text{ or } n \geq 3$$

(A) $n \leq -2$ and $n > 3$

(C) $-2 \leq n < 3$

(B) $n \leq -2$ and $n \geq 3$

(D) $-2 < n \leq 3$

Answer grid for Section I

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

Examination continues overleaf...

Section II

49 marks

Attempt Questions 6 to 10

Allow approximately 84 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 Marks)	Marks
(a) i. Express $-\sqrt{3} + i$ in modulus-argument form. ii. On an Argand diagram, sketch the set of points P that represents the complex number z , which satisfies	1 2
$ z + \sqrt{3} - i = 1$	
iii. Find the set of possible values of $ z $ and $\text{Arg}(z)$, where $\text{Arg}(z)$ is the principal argument.	2
(b) i. On an Argand diagram, draw a neat sketch of	3
$ z - 4 - 4i = 2$ and $\text{Arg}(z) = \frac{\pi}{4}$	
ii. Hence write down all values of z which simultaneously satisfy	2
$ z - 4 - 4i = 2$ and $\text{Arg}(z) = \frac{\pi}{4}$	
(c) The points O , A , B and C on the Argand diagram represent complex numbers 0 , 1 , z and $z + 1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1 and $0 < \theta < \pi$.	
i. With the aid of a diagram or otherwise, explain why $OACB$ is a rhombus.	1
ii. Show that $\frac{z-1}{z+1}$ is purely imaginary.	2
iii. Find the modulus and argument of $z + 1$. Write both the modulus and argument in terms of θ .	2

Question 7 (11 Marks)**Marks**

- (a) Consider the complex polynomial $P(z) = z^4 - 3z^3 + 6z^2 - 12z + 8$. **3**

Given that $P(2i) = 0$, factorise $P(z)$ into linear and quadratic factors with real coefficients.

- (b) i. Show that $(3 - e^{i2\theta})(3 - e^{-i2\theta})$ is purely real. **3**

- ii. Now consider $S = e^{i\theta} + \frac{e^{i3\theta}}{3} + \frac{e^{i5\theta}}{3^2} + \frac{e^{i7\theta}}{3^3} + \dots$. **3**

Show that

$$S = \frac{3(3e^{i\theta} - e^{-i\theta})}{10 - 6\cos 2\theta}$$

- iii. Hence or otherwise, show that **2**

$$\sin \theta + \frac{\sin 3\theta}{3} + \frac{\sin 5\theta}{3^2} + \frac{\sin 7\theta}{3^3} + \dots = \frac{6 \sin \theta}{5 - 3 \cos 2\theta}$$

Question 8 (7 Marks)**Marks**

- (a) Using De Moivre's Theorem, show that **3**

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

- (b) Hence, show that **3**

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

- (c) Briefly explain why **1**

$$\cos 54^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Examination continues overleaf...

Question 9 (7 Marks)**Marks**

- (a) Use mathematical induction to prove for $n \geq 2$, **3**

$$u_n = \frac{1}{2}n(n+1)$$

given $u_1 = 1$ and $u_n = u_{n-1} + n$.

- (b) Suppose $x_i > 0$ for $i = 1, 2, \dots, n$, and $n \geq 2$. **4**

Prove by mathematical induction that

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \geq \frac{n^2}{s}$$

where $s = x_1 + x_2 + \dots + x_n$.

Hint: you may use the fact that $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{c}$ where $a > 0$, $b > 0$ and $c = a + b$, without proving it.

Question 10 (9 Marks)**Marks**

- (a) Prove that $\log_2 6$ is irrational. **2**

- (b) Let P be the proposition

If a pair of angles are alternate and stand on parallel lines, then the pair of angles are equal.

- i. Write the converse of the proposition P , and state whether this is true or not. **2**
- ii. State the contrapositive of P . **1**

- (c) Prove that for all $n \in \mathbb{Z}^+$ and $p \in \mathbb{Z}^+$ where p is prime, that there exists no positive integer m such that **4**

$$(3m+2)^2 = n^2 + p$$

End of paper.

2022 Mathematics Extension 2 Year 12 Course Assessment Task 2 STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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- Q9 - Mathematical Induction

.....

2. Which topics did I need more help with, and what parts specifically?

- Q1-3, 6 - Complex Numbers (Arithmetic, Mod Arg form)

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- Q10 - The Nature of Proof

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3. What other parts from the feedback session can I take away to refine my solutions for future reference?

- Q7 - Complex Numbers (Applications)

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- Q8 - Complex Numbers (De Moivre's Theorem)

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Sample Band E4 Responses

1. (C) 2. (A) 3. (B) 4. (C) 5. (C)

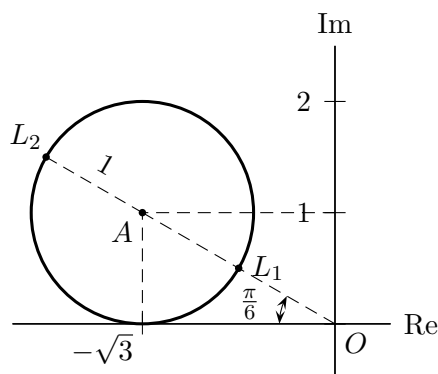
Question 6 (Ham)

(a) i. (1 mark)

$$-\sqrt{3} + i = 2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

(Use diagram to assist with quickly finding the modulus and argument)

ii. (2 marks)



iii. (2 marks)

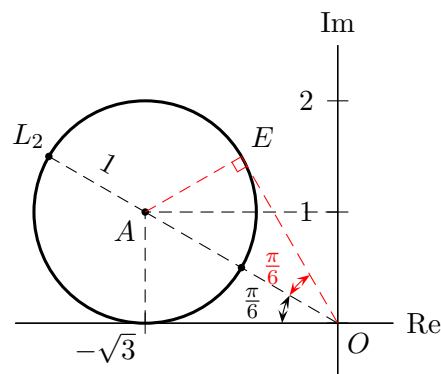
- ✓ [1] for correct range of values for the modulus.
- ✓ [1] for correct range of values for the argument.

Modulus

- $AO = 2$ due to Pythagoras, and with AL_1 being the radius of the circle, hence $OL_1 = 1$
- The furthest point away from O on the circle is the point L_2 , which will have modulus of 3.
- Therefore,

$$1 \leq |z| \leq 3$$

Argument adding more lines to the diagram:



- Construct a new point E on the circle such that $AE \perp OE$.
- Hence the minimum argument for z is

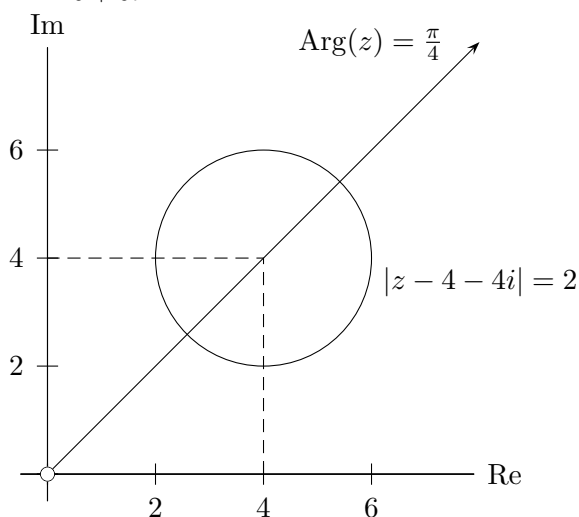
$$\pi - \left(\frac{\pi}{6} \right) - \left(\frac{\pi}{6} \right) = \frac{2\pi}{3}$$

- Max argument for z is π .
- Therefore,

$$\frac{2\pi}{3} \leq \text{Arg}(z) \leq \pi$$

(b) i. (3 marks)

- ✓ [1] for correctly sketching the circle.
- ✓ [1] for correctly sketching the line
- ✓ [1] for open circle on the line at $0 + 0i$



ii. (2 marks)

- ✓ [1] for obtaining a quadratic equation in terms of x only.
- ✓ [1] for both correct x and y values.

$$\begin{cases} y = x & (1) \\ (x - 4)^2 + (y - 4)^2 = 4 & (2) \end{cases}$$

Solving simultaneously by substituting (1) into (2):

$$\begin{aligned}(x-4)^2 + (x-4)^2 &= 4 \\ 2(x-4)^2 &= 4 \\ (x-4)^2 &= 2 \\ \therefore x-4 &= \pm\sqrt{2} \\ x &= 4 \pm \sqrt{2} \\ \therefore y &= 4 \pm \sqrt{2} \\ \therefore z &= (4 \pm \sqrt{2})(1+i)\end{aligned}$$

Alternatively,

$$z-1 = \overrightarrow{AB} \quad z+1 = \overrightarrow{OC}$$

As $\overrightarrow{AB} \perp \overrightarrow{OC}$ due to the diagonals of a rhombus,

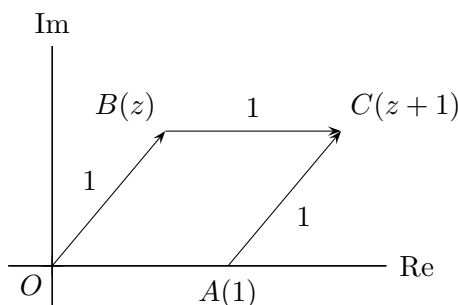
$$\therefore z-1 = \lambda i(z+1)$$

where $\lambda \in \mathbb{R}$, $\lambda \neq 0$. Dividing throughout by $(z+1)$:

$$\therefore \frac{z-1}{z+1} = \lambda i$$

and $\frac{z-1}{z+1}$ is purely imaginary.

(c) i. (1 mark)



- As $z = \cos \theta + i \sin \theta$, $|z| = 1$.
- $|\overrightarrow{OA}| = 1$ as well.
- OB and OA form adjacent sides of a parallelogram $OACB$. However, one pair of adjacent sides are equal.

Hence $OACB$ is a rhombus.

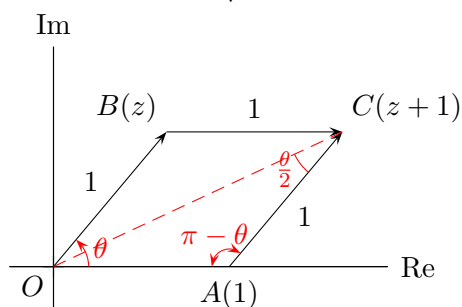
ii. (2 marks)

$$\begin{aligned}& \frac{z-1}{z+1} \\&= \frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1} \\&= \frac{(\cos \theta - 1) + i \sin \theta}{(\cos \theta + 1) + i \sin \theta} \times \frac{[(\cos \theta + 1) - i \sin \theta]}{[(\cos \theta + 1) - i \sin \theta]} \\&= \frac{(\cos \theta - 1)(\cos \theta + 1) - i \sin \theta(\cos \theta - 1) + i \sin \theta(\cos \theta + 1) + \sin^2 \theta}{(\cos \theta + 1)^2 + \sin^2 \theta} \\&= \frac{\cancel{\cos^2 \theta} - 1 + 2i \sin \theta + \cancel{\sin^2 \theta}}{(\cos \theta + 1)^2 + \sin^2 \theta} \\&= i \left(\frac{2 \sin \theta}{(\cos \theta + 1)^2 + \sin^2 \theta} \right)\end{aligned}$$

iii. (2 marks)

✓ [1] for correct expression for $\text{Arg}(z + 1)$

✓ [1] for $|z + 1| = \sqrt{2(1 + \cos \theta)}$



- $\text{Arg}(z + 1) = \frac{\theta}{2}$ as $\text{Arg}(z) = \theta$
(Diagonals of a rhombus bisects the angle)
- As $\triangle AOC$ is isosceles (base side lengths of 1),

$$\therefore \angle OCA = \frac{\theta}{2}$$

and hence

$$\angle OAC = \pi - 2\left(\frac{\theta}{2}\right) = \pi - \theta$$

- Applying the cosine rule in $\triangle OCA$:

$$\begin{aligned} |z + 1|^2 &= 1^2 + 1^2 - 2(1)(1) \cos(\pi - \theta) \\ &= 2 + 2 \cos \theta \end{aligned}$$

$$\begin{aligned} |z + 1| &= \sqrt{2(1 + \cos \theta)} \\ &= \sqrt{2 \times 2 \cos^2 \frac{\theta}{2}} \\ &= 2 \cos \frac{\theta}{2} \end{aligned}$$

Noting that

$$\begin{aligned} \cos^2 \theta &= \frac{1}{2} + \frac{1}{2} \cos 2\theta \\ \cos^2 \frac{\theta}{2} &= \frac{1}{2} + \frac{1}{2} \cos \theta \\ \therefore 2 \cos^2 \frac{\theta}{2} &= 1 + \cos \theta \end{aligned}$$

Alternatively for the modulus:

$$\begin{aligned} |z + 1| &= |(\cos \theta + 1) + i \sin \theta| \\ &= \sqrt{(\cos \theta + 1)^2 + \sin^2 \theta} \\ &= \sqrt{\cos^2 \theta + 2 \cos \theta + 1 + \sin^2 \theta} \\ &= \sqrt{2 \cos \theta + 2} \end{aligned}$$

Question 7 (Ho)

(a) (3 marks)

- ✓ [1] for identifying $-2i$ being a root.
- ✓ [1] for identifying one other factor apart from the complex factors $(z - 2i)$ and $(z + 2i)$.
- ✓ [1] for full factorisation into real factors.

$$P(z) = z^4 - 3z^3 + 6z^2 - 12z + 8$$

$$P(2i) = 0$$

Hence the complex conjugate is also a root, i.e.

$$P(-2i) = 0$$

Hence $(z^2 + 4)$ is a factor

$$\begin{aligned} P(z) &= (z^2 + 0z + 4)(z^2 - 3z + 2) \\ &= (z^2 + 4)(z - 2)(z - 1) \end{aligned}$$

(b) i. (3 marks)

- ✓ [1] for correct expansion to

$$9 - (3e^{i2\theta} + 3e^{-i2\theta}) + e^0$$

- ✓ [1] for correct simplification of complex exponential terms to $6 \cos 2\theta$
- ✓ [1] for recognition that the final expression does not contain any imaginary part.

$$\begin{aligned} &(3 - e^{i2\theta})(3 - e^{-i2\theta}) \\ &= 9 - (3e^{i2\theta} + 3e^{-i2\theta}) + e^0 \\ &= 10 - 3(e^{i2\theta} + e^{-i2\theta}) \\ &= 10 - 3 \times 2 \cos 2\theta \\ &= 10 - 6 \cos 2\theta \\ &\quad \text{Im}(10 - 6 \cos 2\theta) = 0 \end{aligned}$$

Hence, $(3 - e^{i2\theta})(3 - e^{-i2\theta})$ is purely real.

ii. (3 marks)

- ✓ [1] for recognising $|r| < 1$

- ✓ [1] for substituting into limiting sum formula and multiplying by the conjugate $(3 - e^{-i2\theta})$

- ✓ [1] for showing the final result.

As $r = \frac{1}{3}e^{i2\theta}$, $|r| = \left|\frac{1}{3}e^{i2\theta}\right| < 1$ given $|e^{i2\theta}| = 1$

$$\begin{aligned} S &= \frac{a}{1 - r} \\ &= \frac{e^{i\theta}}{1 - \frac{1}{3}e^{i2\theta}} \times \frac{3}{3} \\ &= \frac{3e^{i\theta}}{3 - e^{i2\theta}} \times \frac{(3 - e^{-i2\theta})}{(3 - e^{-i2\theta})} \\ &= \frac{3e^{i\theta}(3 - e^{-i2\theta})}{3(3 - e^{i2\theta} - e^{-i2\theta})} \\ &= \frac{3e^{i\theta}(3 - e^{-i2\theta})}{10 - 6 \cos 2\theta} \\ &= \frac{3(3e^{i\theta} - e^{-i\theta})}{10 - 6 \cos 2\theta} \end{aligned}$$

iii. (2 marks)

- ✓ [1] for recognising the imaginary part of the summation from the previous part is required.
- ✓ [1] for final required proof.

$$\sin \theta = \text{Im}(e^{i\theta})$$

$$\sin 3\theta = \text{Im}(e^{i3\theta})$$

etc. Hence, the required summation is simply

$$\begin{aligned} &\sin \theta + \frac{\sin 3\theta}{3} + \frac{\sin 5\theta}{3^2} + \dots \\ &= \text{Im} \left(\frac{3(3e^{i\theta} - e^{-i\theta})}{10 - 6 \cos 2\theta} \right) \\ &= \frac{3}{10 - 6 \cos 2\theta} (3 \sin \theta - \sin(-\theta)) \\ &= \frac{3(3 \sin \theta + \sin \theta)}{10 - 6 \cos 2\theta} \\ &= \frac{12 \sin \theta}{10 - 6 \cos 2\theta} \\ &= \frac{6 \sin \theta}{5 - 3 \cos 2\theta} \end{aligned}$$

Question 8 (Ho)

(a) (3 marks)

- ✓ [1] for correct usage of De Moivre's Theorem and expansion via the Binomial Theorem
- ✓ [1] for correct reduction of $\sin \theta$ terms into $\cos \theta$
- ✓ [1] for final result.

$$\begin{aligned}
 \cos 5\theta + i \sin 5\theta &= (\cos \theta + i \sin \theta)^5 \\
 &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta \\
 &\quad + i^2 10 \cos^3 \theta \sin^2 \theta \\
 &\quad + i^3 10 \cos^2 \theta \sin^3 \theta \\
 &\quad + i^4 5 \cos \theta \sin^4 \theta \\
 &\quad + i^5 \sin^5 \theta
 \end{aligned}$$

Equating real parts,

$$\begin{aligned}
 \cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\
 &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) \\
 &\quad + 5 \cos \theta (1 - \cos^2 \theta)^2 \\
 &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta \\
 &\quad + 5 \cos \theta (1 - 2 \cos^2 \theta + \cos^4 \theta) \\
 &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta \\
 &\quad + 5 \cos \theta - 10 \cos^3 \theta + 5 \cos^5 \theta \\
 &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta
 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for obtaining quadratic in terms of m .
- ✓ [1] for obtaining $\cos \theta = \pm \sqrt{\frac{10 \pm 2\sqrt{5}}{16}}$.
- ✓ [1] for final result.
- $\cos(5 \times 18^\circ) = \cos 90^\circ = 0$
- Solve $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta = 0$ to obtain the surd expression to $\cos 18^\circ$, i.e.

$$\begin{aligned}
 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta &= 0 \\
 \cos \theta (16 \cos^4 \theta - 20 \cos^2 \theta + 5) &= 0
 \end{aligned}$$

As $\cos 18^\circ \neq 0$,

$$\therefore 16 \cos^4 \theta - 20 \cos^2 \theta + 5 = 0$$

Let $m = \cos^2 \theta$,

$$16m^2 - 20m + 5 = 0$$

Applying the quadratic formula,

$$\begin{aligned}
 m &= \frac{20 \pm \sqrt{20^2 - 4(16)(5)}}{2(16)} \\
 &= \frac{20 \pm \sqrt{80}}{32} \\
 &= \frac{20 \pm 4\sqrt{5}}{32} \\
 &= \frac{10 \pm 2\sqrt{5}}{16} \\
 \therefore \cos^2 \theta &= \frac{10 \pm 2\sqrt{5}}{16} \\
 \cos \theta &= \pm \sqrt{\frac{10 \pm 2\sqrt{5}}{16}}
 \end{aligned}$$

As 18° is in the first quadrant and $\cos \theta > 0$ for first quadrant values of θ , the negative root is discarded.

$$\therefore \cos \theta = \frac{\sqrt{10 \pm 2\sqrt{5}}}{4}$$

Also, $\cos 18^\circ$ is closer to $\cos 0^\circ = 1$, take the value which is closer to 1:

$$\therefore \cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

as $\frac{\sqrt{10 - 2\sqrt{5}}}{4}$ is closer to 0.

(c) (1 mark)

$$\cos(5 \times 54^\circ) = \cos 270^\circ = 0$$

No other acute angle θ would result in $\cos 5\theta = 0$, and from the graph of $y = \cos x$, $\cos 54^\circ < \cos 18^\circ$.

Question 9 (Sekaran)

(a) (3 marks)

- ✓ [1] for correct testing of the base case.
- ✓ [1] for correct usage of the inductive hypothesis in $P(k+1)$
- ✓ [1] for final result.

Let $P(n)$ be the proposition

$$u_n = \frac{1}{2}n(n+1)$$

where $u_1 = 1, u_n = u_{n-1} + n$

1. Base case: $P(2)$

$$u_2 = u_1 + 2 = 1 + 2 = 3$$

Also,

$$u_2 = \frac{1}{2}(2)(2+1) = 3$$

Hence $P(2)$ is true.

2. Inductive step: assume $P(k)$ is true, i.e.

$$u_k = \frac{1}{2}k(k+1)$$

$$\text{where } u_1 = 1, u_k = u_{k-1} + k$$

Examine $P(k+1)$:

$$\begin{aligned} u_{k+1} &= u_k + (k+1) \\ &= \frac{1}{2}k(k+1) + (k+1) \\ &= \frac{1}{2}(k+1)(k+2) \\ &= \frac{1}{2}(k+1)((k+1)+1) \end{aligned}$$

- (b) (4 marks)

- ✓ [1] for correct testing of the base case.
- ✓ [1] for correct usage of the inductive hypothesis in $P(k+1)$
- ✓ [1] for further testing of $P(k+1)$
- ✓ [1] for final result.

Let $P(n)$ be the proposition

$$\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \geq \frac{n^2}{s}$$

where $s = x_1 + x_2 + \cdots + x_n$, $n \geq 2$.

1. Base case: $P(2)$

- Left side: $\frac{1}{x_1} + \frac{1}{x_2}$
- Right: $\frac{2^2}{x_1 + x_2} = \frac{4}{x_1 + x_2}$.

$P(2)$ is true by the identity provided,
 $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$.

2. Inductive step: assume $P(k)$ is true, i.e.

$$\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_k} \geq \frac{k^2}{x_1 + x_2 + \cdots + x_k}$$

Examine $P(k+1)$:

$$\begin{aligned} &\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \\ &\geq \frac{k^2}{x_1 + x_2 + \cdots + x_k} + \frac{1}{x_{k+1}} \end{aligned}$$

Note: the requirement is to prove

$$\begin{aligned} &\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \\ &\geq \frac{(k+1)^2}{x_1 + x_2 + \cdots + x_k + x_{k+1}} \end{aligned}$$

Now check for whether

$$\frac{k^2}{x_1 + x_2 + \cdots + x_k} + \frac{1}{x_{k+1}} \quad (\dagger)$$

is greater than

$$\frac{(k+1)^2}{x_1 + x_2 + \cdots + x_k + x_{k+1}} \quad (\ddagger)$$

or not: complete a subtraction of $(\ddagger)$ from $(\dagger)$ and check whether that is greater than zero.

$$\begin{aligned} &\frac{k^2}{x_1 + x_2 + \cdots + x_k} + \frac{1}{x_{k+1}} \\ &\quad - \frac{(k+1)^2}{x_1 + x_2 + \cdots + x_k + x_{k+1}} \\ &= \frac{k^2}{a} + \frac{1}{b} - \frac{(k+1)^2}{a+b} \end{aligned}$$

where

$$a = x_1 + x_2 + \cdots + x_k$$

$$b = x_{k+1}$$

$$\begin{aligned} &\frac{k^2}{a} + \frac{1}{b} - \frac{(k+1)^2}{a+b} \\ &= \frac{k^2b(b+a) + a(a+b) - (k+1)^2ab}{ab(a+b)} \\ &= \frac{k^2b^2 + k^2ab + a^2 + ab - k^2ab - 2kab - ab}{ab(a+b)} \\ &= \frac{k^2b^2 - 2kab + a^2}{ab(a+b)} \\ &= \frac{(kb - a)^2}{ab(a+b)} \\ &\geq 0 \end{aligned}$$

Hence

$$\begin{aligned} \frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_k} + \frac{1}{x_{k+1}} \\ \geq \frac{k^2}{x_1 + x_2 + \cdots + x_k} + \frac{1}{x_{k+1}} \\ \geq \frac{(k+1)^2}{x_1 + x_2 + \cdots + x_k + x_{k+1}} \end{aligned}$$

which makes $P(k+1)$ true, and $P(n)$ true by induction.

Question 10 (Sekaran)

(a) (2 marks)

- ✓ [1] for setting up the contradiction.
- ✓ [1] for final proof.

Two of the techniques are presented here.

Method 1

- Assume $\log_2 6$ is rational, i.e.

$$\log_2 6 = \frac{p}{q}$$

where $p, q \in \mathbb{Z}^+$, and p and q are coprime such that $p \neq q$

- Then,

$$\begin{aligned} 2^{\frac{p}{q}} &= 6 \\ 2^p &= 6^q \\ 2^p &= 2^q 3^q \\ \frac{2^p}{2^q} &= 3^q \end{aligned}$$

- 3 divides $2^p 3^q$. Hence, 3 divides 2^p , which is a contradiction as 2^p is an even number.
- Hence $2^{\frac{p}{q}} \neq 6$, and $\log_2 6 \neq \frac{p}{q}$ and $\log_2 6$ is irrational.

Method 2

$$\frac{p}{q} = \log_2 6 > \log_2 4 > 2$$

$$\therefore \frac{p}{-q} > \frac{2q}{-q}$$

$$p - q > q$$

Note also that $p - q \in \mathbb{Z}^+$ as $q \in \mathbb{Z}^+$. Hence,

$$\begin{aligned} 2^p &= 2^q 3^q \\ \therefore 2^{p-q} &= 3^q \end{aligned}$$

which is a contradiction as 2^{p-q} is even and 3^q is odd.

(b) i. (2 marks) Let

- p be the statement *a pair of angles are alternate*
- q be the statement *stand on parallel lines*
- r be the statement *pair of angles are equal*

Then the statement becomes

$$(p \wedge q) \Rightarrow r$$

The converse is thus

$$r \Rightarrow (p \wedge q)$$

i.e.

If the pair of angles are equal, then the angles are alternate *and* stand on parallel lines.

This is not necessarily true.

ii. (1 mark) The contrapositive is:

$$\neg r \Rightarrow \neg(p \wedge q)$$

Note that

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

i.e.

If a pair of angles are not equal, then the angles are not alternate or not stand on parallel lines.

(c) (4 marks)

- ✓ [1] Factorises expression for p .
- ✓ [1] Realises the factors must be 1 and p .
- ✓ [1] for further substantial progress
- ✓ [1] for final proof.
- Let m, n and p be positive integers where p is prime, and $(3m+2)^2 = n^2 + p$.
- Then,

$$\begin{aligned} p &= (3m+2)^2 - n^2 \\ &= (3m+2-n)(3m+2+n) \end{aligned}$$

- If m , n , and p are positive integers, then

$$0 < (3m + 2 - n) < (3m + 2 + n)$$

- Also given p is prime, then

$$(3m + 2 - n) = 1 \quad (10.1)$$

$$(3m + 2 + n) = p \quad (10.2)$$

as it contains only two factors.

- Adding (10.1) and (10.2):

$$6m + 4 = p + 1$$

$$\therefore p = 6m + 3 = 3(2m + 1)$$

which is a contradiction: p is meant to be prime, but the result indicates p is also divisible by 3. Hence for all integers n , p , where p is prime, there is no positive integer m such that

$$(3m + 2)^2 = n^2 + p$$